

Zeynep Doğa Dellal

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EDUCATION

Technische Hochschule Deggendorf

Master of Engineering in Artificial Intelligence for Smart Sensors and Actuators

Bilkent University

Bachelor of Science in Computer Engineering

Cham, Germany

Mar 2026 – Present

Ankara, Turkey

Sep 2020 – Jun 2025

EXPERIENCE

Embedded Systems Intern

Aug 2024 – Sep 2024

Karel

Ankara, Turkey

- Contributed to GENTSIS by designing PCBs, integrating sensors, and optimizing ESP32-based IoT communication performance.
- Processed and structured 1M+ transport records using Python and MySQL for routing optimization and analysis, reducing processing time by 20%.

Undergraduate Assistant

Oct 2023 – May 2024

Bilkent University

Ankara, Turkey

- Assisted Prof. Robin Ann Downey in organizing the Responsible Artificial Intelligence Awards with MathWorks and received acknowledgment in her [SEFI-2024 publication](#) for this work.

Software Developer

Sep 2023 – Jan 2024

ForyCode Studios

İstanbul, Turkey

- Built and integrated 10+ gameplay mechanics in Unity using C#, including player control, combat, and level systems.

Backend Development Intern

Jun 2023 – Jul 2023

Medyasoft

İstanbul, Turkey

- Developed backend modules and APIs using C#, optimizing database logic to improve system performance.

Android Developer

Nov 2021 – Nov 2022

BenzinLitre

İstanbul, Turkey

- Developed and optimized 15+ features in Kotlin/Java using MVVM architecture in Android Studio.

PROJECTS

Drunkenness Detection System | *Python, XGBoost, OpenCV, Raspberry Pi*

- Developed a multi-sensor driver impairment detection system combining alcohol, eye tracking, and heart-rate data.
- Trained an XGBoost model and used SHAP for feature importance analysis.

Veritas (AI News App) | *PyTorch, spaCy, FastAPI, Django, PostgreSQL, React Native*

- Built and deployed a cross-platform news app integrating custom LLMs for unbiased content generation and analysis.
- Deployed on a self-managed server.

Aircraft Detection System | *PyTorch, Hugging Face, YOLOv8*

- Developed and trained custom CNN, ResNet, and hybrid models, including RCNN and YOLOv8, for aircraft detection and bounding box regression in satellite images, achieving 92% accuracy.

TECHNICAL SKILLS

Programming: Python, C, C++, C#, Java, JavaScript, TypeScript, Assembly (ARM/x86), Verilog

Embedded Systems: ESP32, ESP8266, Raspberry Pi, ESP-IDF, Zephyr RTOS, Basys 3 FPGA, Vivado, BME280, MQ-3, IR Pulse Sensor, Pi NoIR Camera

AI/ML: PyTorch, XGBoost, YOLOv8, ResNet, CNNs, Hugging Face, OpenAI API, SHAP

Tools: Git, Linux, Docker, Jupyter Notebook, Jira

Backend & APIs: FastAPI, Django, ASP.NET MVC, REST APIs, PostgreSQL, MySQL

Languages: English (IELTS 8.0), Turkish (Native), German (A2)